Exchangeable Dyads: Model and Back-Transformation

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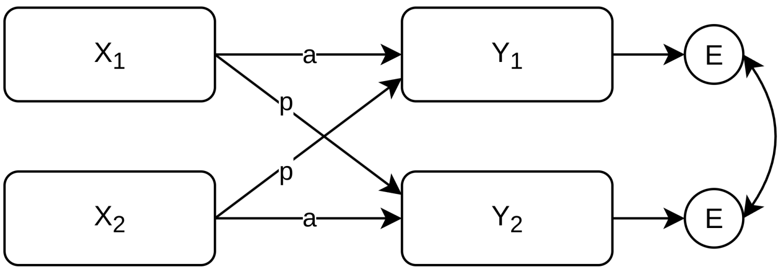
Full GitHub repository: The full repository contains the tutorial source files, reusable helper functions, model code, rendered PDFs, and additional information: https://github.com/Pascal-Kueng/05DyadicDataAnalysis

1 Background

Exchangeable dyads do not have substantively meaningful partner roles. Partner labels are arbitrary, and flipping the labels should not change the model interpretation.

For intensive longitudinal data, an exchangeable APIM needs to model:

- pooled fixed effects, because the two partner labels are not substantively different;
- pooled average time trends and pooled actor/partner effects;
- stable between-dyad differences in average levels and slopes;
- partner deviations from those dyad-level effects, because the two people in a dyad are still distinct repeated-observation units;
- correlations among the dyad-level random effects and among the partner deviation random effects;
- same-day residual interdependence between partners after fixed and random effects have been accounted for;
- a residual structure that remains invariant when the arbitrary partner labels are switched.



Assume `df_long_dim` contains one row per person-day and two rows per `coupleID:diaryday`. The exchangeability code uses `Idiff = +1/-1` to represent partner deviations from the dyad-level mean.

2 Data structure

Table 1: Example person-day structure for an exchangeable dyad

coupleID	userID	diaryday	couple_day	member	Idiff	is_partner1	is_partner2
31	31_1	0	31.0	1	1	1	0
31	31_1	1	31.1	1	1	1	0
31	31_1	2	31.2	1	1	1	0
31	31_2	0	31.0	2	-1	0	1
31	31_2	1	31.1	2	-1	0	1
31	31_2	2	31.2	2	-1	0	1

`Idiff` is constant within person and opposite within dyad (randomly assigned). The dummy variables used in `glmmTMB`, `is_partner1` and `is_partner2`, must be constructed from the same `Idiff` coding (e.g. `is_partner1 = 1` when `Idiff == 1` and `is_partner2 = 1` when `Idiff == -1`.)

3 glmmTMB model

```
model_exch_long_apim_glmmtmb <- glmmTMB(
  closeness ~ 1 +

    # Pooled fixed effects: no gender-specific intercepts or slopes.
    diaryday_c +
    provided_support_actor_cwp +
    provided_support_partner_cwp +
    provided_support_actor_cbp +
    provided_support_partner_cbp +

    # Dyad-level means and slopes.
    (1 + diaryday_c +
       provided_support_actor_cwp +
       provided_support_partner_cwp | coupleID) +

    # Partner deviations from the dyad-level effects.
    # This block is kept separate from the dyad-level block so that the
    # sum and difference components are independent.
    (0 + Idiff +
       I(Idiff * diaryday_c) +
       I(Idiff * provided_support_actor_cwp) +
       I(Idiff * provided_support_partner_cwp) | coupleID) +

    # Same-day residual interdependence:
    # homogeneous compound symmetry = one residual variance plus one
    # partner residual correlation.
    homcs(0 + is_partner1 + is_partner2 | coupleID:diaryday),

  # The residual variance is represented by the homcs block above.
  # Without dispformula = ~ 0, glmmTMB would add a second independent
  # residual variance on top of the dyadic residual covariance structure.
  # For families without a residual, dispformula should NOT be set to zero.
  dispformula = ~ 0,
  family = gaussian(),
  data = df_long_dim
)
```

The residual block could also be written as independent sum/difference terms:

```
(1 | coupleID:diaryday) +
(0 + Idiff | coupleID:diaryday)
```

With `Idiff = +1/-1`, this has the same parameter count as `homcs(...)`: one sum variance and one difference variance, which rotate back to one homogeneous residual variance and one residual covariance.

4 brms model

```
formula_exch_long_apim <- bf(
  closeness ~ 1 +

    # Pooled fixed effects.
    diaryday_c +
    provided_support_actor_cwp +
    provided_support_partner_cwp +
    provided_support_actor_cbp +
    provided_support_partner_cbp +

    # Dyad-level means and slopes.
    (1 + diaryday_c +
       provided_support_actor_cwp +
       provided_support_partner_cwp | coupleID) +

    # Partner deviations from the dyad-level effects.
    (0 + Idiff +
       I(Idiff * diaryday_c) +
       I(Idiff * provided_support_actor_cwp) +
       I(Idiff * provided_support_partner_cwp) | coupleID) +

    # Native residual correlation between the two partners on the same day.
    # We use unstr() instead of cs() because brms currently restricts cs()
    # residual correlations to be positive. With exactly two members per group
    # (dyads), unstr() is equivalent here because only one residual correlation
    # is estimated. For triads or larger groups, unstr() would no longer impose
    # the compound-symmetry constraint.
    unstr(time = member, gr = couple_day)

  # No sigma formula is specified here. Thus, sigma ~ 1 is implied,
  # meaning one homogeneous residual variance for both partners.
  # If sigma is modelled with its own formula, brms estimates the sigma
  # coefficients on the log scale, so those coefficients must be exponentiated
  # before reporting them as residual SDs.
)

priors_exch_long_apim <- c(
  prior(normal(4, 2), class = "Intercept"),
  prior(normal(0, 2), class = "b"),
  prior(exponential(1), class = "sd"),
  prior(lkj(2), class = "cor"),
  prior(student_t(3, 0, 1.5), class = "sigma"),
  prior(lkj(2), class = "cortime")
)

model_exch_long_apim <- brm(
  formula = formula_exch_long_apim,
  data = df_long_dim,
  family = gaussian(link = identity),
  prior = priors_exch_long_apim,
  chains = 4,
  cores = 4,
  iter = 2000,
  warmup = 1000,
  seed = 123,
  file = file.path("brms_cache", "model_apim_ind_long_apim")
)
```

5 Back-transformation

The model estimates a sum block and an `Idiff` block. For exchangeable dyads, the full APIM random-effects matrix is recovered by rotating these two blocks:

$$\mathrm{Var}(A) = \mathrm{Var}(B) = \Sigma_{\text{sum}} + \Sigma_{\text{diff}}$$

$$\mathrm{Cov}(A, B) = \Sigma_{\text{sum}} - \Sigma_{\text{diff}}$$

This exact rotation assumes that the sum and difference random-effect blocks are independent.

The helper function `summarize_exchangeable_apim_brms()` used below is defined in `00_R_Functions/ReportModels.R` on GitHub. It reconstructs the sum and `Idiff` covariance matrices for every posterior draw, rotates each draw into the full APIM partner-level matrix, and then summarizes the transformed covariance, SD, correlation, fixed-effect, and residual quantities.

For `glmmTMB`, the same back-transformation is algebraically just as straightforward, but interval estimation (CI) is less direct because there are no posterior draws. A helper function implementing the delta-method is also available on GitHub.

```
apim_results <- summarize_exchangeable_apim_brms(
  model_exch_long_apim,
  gr = "coupleID",
  idiff = "Idiff",
  term_labels = c("int", "day", "actor_wp", "partner_wp"),
  partner_labels = c("A", "B")
)
```

6 Variance-covariance matrix

```
kable(
  round(apim_results$full_covariance_matrix, 3),
  caption = "Back-transformed full APIM random-effects variance-covariance matrix",
  booktabs = TRUE
) |>
  kable_styling(latex_options = "scale_down", font_size = 7)
```

Table 2: Back-transformed full APIM random-effects variance-covariance matrix

	A_int	A_day	A_actor_wp	A_partner_wp	B_int	B_day	B_actor_wp	B_partner_wp
A_int	1.176	0.011	0.095	0.018	-0.275	-0.002	-0.064	0.004
A_day	0.011	0.000	0.001	0.000	-0.002	0.000	-0.001	0.000
A_actor_wp	0.095	0.001	0.024	0.004	-0.064	-0.001	-0.018	-0.002
A_partner_wp	0.018	0.000	0.004	0.012	0.004	0.000	-0.002	0.003
B_int	-0.275	-0.002	-0.064	0.004	1.176	0.011	0.095	0.018
B_day	-0.002	0.000	-0.001	0.000	0.011	0.000	0.001	0.000
B_actor_wp	-0.064	-0.001	-0.018	-0.002	0.095	0.001	0.024	0.004
B_partner_wp	0.004	0.000	-0.002	0.003	0.018	0.000	0.004	0.012

7 SD table

```r
kable(
  apim_results$sd_summary,
  digits = 3,
  caption = "Back-transformed APIM random-effect standard deviations with 95% credible intervals"
)
```

Table 3: Back-transformed APIM random-effect standard deviations with 95% credible intervals

parameter	estimate	est_error	q_low	q_high
A_actor_wp	0.153	0.018	0.119	0.190
A_day	0.018	0.001	0.016	0.020
A_int	1.083	0.057	0.976	1.196
A_partner_wp	0.107	0.021	0.065	0.147
B_actor_wp	0.153	0.018	0.119	0.190
B_day	0.018	0.001	0.016	0.020
B_int	1.083	0.057	0.976	1.196
B_partner_wp	0.107	0.021	0.065	0.147

8 Correlation matrix

```r
kable(
  round(apim_results$full_correlation_matrix, 3),
  caption = "Back-transformed APIM random-effect correlation matrix",
  booktabs = TRUE
) |>
  kable_styling(latex_options = "scale_down", font_size = 7)
```

Table 4: Back-transformed APIM random-effect correlation matrix

	A_int	A_day	A_actor_wp	A_partner_wp	B_int	B_day	B_actor_wp	B_partner_wp
A_int	1.000	0.579	0.569	0.153	-0.233	-0.128	-0.380	0.038
A_day	0.579	1.000	0.362	0.144	-0.128	-0.134	-0.207	0.045
A_actor_wp	0.569	0.362	1.000	0.253	-0.380	-0.207	-0.757	-0.105
A_partner_wp	0.153	0.144	0.253	1.000	0.038	0.045	-0.105	0.241
B_int	-0.233	-0.128	-0.380	0.038	1.000	0.579	0.569	0.153
B_day	-0.128	-0.134	-0.207	0.045	0.579	1.000	0.362	0.144
B_actor_wp	-0.380	-0.207	-0.757	-0.105	0.569	0.362	1.000	0.253
B_partner_wp	0.038	0.045	-0.105	0.241	0.153	0.144	0.253	1.000

9 Final reporting summary

Table 5: Compact reporting summary

section	parameter	estimate	est_error	q_low	q_high
Fixed effects	Intercept	5.111	0.064	4.990	5.235
Fixed effects	diaryday_c	0.004	0.001	0.001	0.006
Fixed effects	provided_support_actor_cwp	0.300	0.013	0.274	0.327
Fixed effects	provided_support_partner_cwp	0.149	0.015	0.120	0.179

Table 5: Compact reporting summary *(continued)*

section	parameter	estimate	est_error	q_low	q_high
Fixed effects	provided_support_actor_cbp	1.304	0.088	1.136	1.477
Fixed effects	provided_support_partner_cbp	0.402	0.082	0.232	0.560
Back-transformed random-effect SDs	A_actor_wp	0.153	0.018	0.119	0.190
Back-transformed random-effect SDs	A_day	0.018	0.001	0.016	0.020
Back-transformed random-effect SDs	A_int	1.083	0.057	0.976	1.196
Back-transformed random-effect SDs	A_partner_wp	0.107	0.021	0.065	0.147
Back-transformed random-effect SDs	B_actor_wp	0.153	0.018	0.119	0.190
Back-transformed random-effect SDs	B_day	0.018	0.001	0.016	0.020
Back-transformed random-effect SDs	B_int	1.083	0.057	0.976	1.196
Back-transformed random-effect SDs	B_partner_wp	0.107	0.021	0.065	0.147
Back-transformed random-effect correlations	A_actor_wp with A_day	0.364	0.102	0.150	0.550
Back-transformed random-effect correlations	A_actor_wp with A_int	0.572	0.085	0.386	0.722
Back-transformed random-effect correlations	A_day with A_int	0.579	0.054	0.467	0.678
Back-transformed random-effect correlations	A_partner_wp with A_actor_wp	0.265	0.157	-0.026	0.574
Back-transformed random-effect correlations	A_partner_wp with A_day	0.147	0.130	-0.112	0.399
Back-transformed random-effect correlations	A_partner_wp with A_int	0.157	0.119	-0.077	0.387
Back-transformed random-effect correlations	B_actor_wp with A_actor_wp	-0.759	0.153	-0.991	-0.414
Back-transformed random-effect correlations	B_actor_wp with A_day	-0.209	0.112	-0.418	0.024
Back-transformed random-effect correlations	B_actor_wp with A_int	-0.382	0.102	-0.573	-0.182
Back-transformed random-effect correlations	B_actor_wp with A_partner_wp	-0.110	0.159	-0.414	0.199
Back-transformed random-effect correlations	B_actor_wp with B_day	0.364	0.102	0.150	0.550
Back-transformed random-effect correlations	B_actor_wp with B_int	0.572	0.085	0.386	0.722
Back-transformed random-effect correlations	B_day with A_actor_wp	-0.209	0.112	-0.418	0.024
Back-transformed random-effect correlations	B_day with A_day	-0.133	0.120	-0.364	0.105
Back-transformed random-effect correlations	B_day with A_int	-0.128	0.085	-0.291	0.039
Back-transformed random-effect correlations	B_day with A_partner_wp	0.046	0.135	-0.214	0.305
Back-transformed random-effect correlations	B_day with B_int	0.579	0.054	0.467	0.678
Back-transformed random-effect correlations	B_int with A_actor_wp	-0.382	0.102	-0.573	-0.182
Back-transformed random-effect correlations	B_int with A_day	-0.128	0.085	-0.291	0.039
Back-transformed random-effect correlations	B_int with A_int	-0.232	0.091	-0.405	-0.054
Back-transformed random-effect correlations	B_int with A_partner_wp	0.039	0.126	-0.207	0.279
Back-transformed random-effect correlations	B_partner_wp with A_actor_wp	-0.110	0.159	-0.414	0.199
Back-transformed random-effect correlations	B_partner_wp with A_day	0.046	0.135	-0.214	0.305
Back-transformed random-effect correlations	B_partner_wp with A_int	0.039	0.126	-0.207	0.279
Back-transformed random-effect correlations	B_partner_wp with A_partner_wp	0.263	0.381	-0.522	0.972
Back-transformed random-effect correlations	B_partner_wp with B_actor_wp	0.265	0.157	-0.026	0.574
Back-transformed random-effect correlations	B_partner_wp with B_day	0.147	0.130	-0.112	0.399
Back-transformed random-effect correlations	B_partner_wp with B_int	0.157	0.119	-0.077	0.387
Residual structure	residual_variance	0.878	0.012	0.855	0.902
Residual structure	same_day_residual_correlation	0.038	0.014	0.008	0.064
Residual structure	same_day_residual_covariance	0.033	0.013	0.007	0.056
Residual structure	sigma	0.937	0.007	0.925	0.950